Mecanum Robot Movement Calibration R&D; Template

This template is intended for calibrating PWM values to minimize movement offset. Leave the PWM values and target blank until the desired test parameters are determined.

Common Test Conditions

- Use the same starting position and orientation for every trial.
- Use the same testing surface.
- Battery should be fully charged before testing.
- Perform 5 trials for each experiment.
- Measure the final movement offset after reaching the target.
- Calculate the average offset.

Forward

Function: **forward(target)**

Experiment 1: Set all motor PWM values equal.

Experiment 2: Set left-side PWM higher than right-side PWM.

Experiment 3: Set left-side PWM lower than right-side PWM.

Experiment	UL PWM	LL PWM	UR PWM	LR PWM	Target	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average Offset
1											
2											
3											

Observation: ____

Conclusion / Selected PWM: ________________________________

Backward

Function: **moveback(target)**

Experiment 1: Set all motor PWM values equal.

Experiment 2: Set left-side PWM higher than right-side PWM.

Experiment 3: Set left-side PWM lower than right-side PWM.

Experiment	UL PWM	LL PWM	UR PWM	LR PWM	Target	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average Offset
1											
2											
3											

Observation: ____

Conclusion / Selected PWM: ________________________________

Left Strafe

Function: **moveleft(target)**

Experiment 1: Set all motor PWM values equal.

Experiment 2: Set left-side PWM higher than right-side PWM.

Experiment 3: Set left-side PWM lower than right-side PWM.

Experiment	UL PWM	LL PWM	UR PWM	LR PWM	Target	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average Offset
1											
2											
3											

Observation: ____

Conclusion / Selected PWM: _____________________________________

Right Strafe

Function: **moveright(target)**

Experiment 1: Set all motor PWM values equal.

Experiment 2: Set left-side PWM higher than right-side PWM.

Experiment 3: Set left-side PWM lower than right-side PWM.

Experiment	UL PWM	LL PWM	UR PWM	LR PWM	Target	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average Offset
1											
2											
3											

Observation: ____

Conclusion / Selected PWM: _____________________________________

Diagonal Upper Right

Function: **diagonalUpperRight(target)**

Experiment 1: Set all motor PWM values equal.

Experiment 2: Set left-side PWM higher than right-side PWM.

Experiment 3: Set left-side PWM lower than right-side PWM.

Experiment	UL PWM	LL PWM	UR PWM	LR PWM	Target	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average Offset
1											

2											
3											

Observation: ____

Conclusion / Selected PWM: _________________________________

Diagonal Upper Left

Function: **diagonalUpperLeft(target)**

Experiment 1: Set all motor PWM values equal.

Experiment 2: Set left-side PWM higher than right-side PWM.

Experiment 3: Set left-side PWM lower than right-side PWM.

Experiment	UL PWM	LL PWM	UR PWM	LR PWM	Target	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average Offset
1											
2											
3											

Observation: ____

Conclusion / Selected PWM: _________________________________

Diagonal Lower Right

Function: **diagonalLowerRight(target)**

Experiment 1: Set all motor PWM values equal.

Experiment 2: Set left-side PWM higher than right-side PWM.

Experiment 3: Set left-side PWM lower than right-side PWM.

Experiment	UL PWM	LL PWM	UR PWM	LR PWM	Target	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average Offset
1											
2											
3											

Observation: ____

Conclusion / Selected PWM: _________________________________

Diagonal Lower Left

Function: **diagonalLowerLeft(target)**

Experiment 1: Set all motor PWM values equal.

Experiment 2: Set left-side PWM higher than right-side PWM.

Experiment 3: Set left-side PWM lower than right-side PWM.

Experiment	UL PWM	LL PWM	UR PWM	LR PWM	Target	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average Offset
1											
2											
3											

Observation: ____

Conclusion / Selected PWM: ___________________________________